

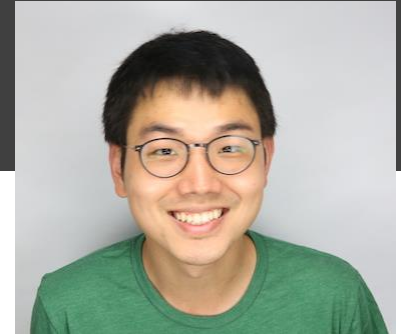
DSC 102

Systems for Scalable Analytics

Fall 2026

Haojian Jin

About Myself



Haojian Jin (<http://haojianj.in/>)

Asst. Prof @ UCSD-HDSI

Ph.D. from CMU Human-Computer Interaction Institute

Ph.D. Thesis: Modular Privacy Flow

Before Ph.D.: worked at Yahoo Research, ran a startup
HCI, Software Engineering, Mobile Computing, AI.



The best way to predict the future is
to invent it.

— *Alan Kay* —

AZ QUOTES

What is this course about? Why take it?



Reddit · r/UCSD

10+ comments · 2 years ago

DSC 102 in a nutshell. : r/UCSD



IVEBEENGRAPED · 3y ago

This class was honestly the most useful class I took at UCSD. I'm in my first job out of school, and half of what I do here is messing around with AWS and Spark like we did in that class. Would highly recommend, even to non-DS majors.



19



Reply



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atvrider512 · 3y ago

yeah lol this was me, I feel like this material is sooooo useful but the class was so disorganized I didn't get to fully learn and process it



2



Reply



Share



phatfat · 3y ago

this class hurt me



2



Reply



Share



https://www.reddit.com/r/UCSD/comments/npqcdm/dsc_102_in_a_nutshell/

How much does a Statistician make?

Updated Jan 4, 2022

Industry

All industries ▾

Employer Size

All company sizes ▾

Experience

All years of Experience ▾

 Very High Confidence

\$88,989 /yr

Average Base Pay

2,398 salaries





Data Scientist Salaries United States ▼

Overview

Salaries

Interviews

Insights

Career Path

How much does a Data Scientist make?

Updated Jan 4, 2022

Industry



All industries



Employer Size



All company sizes



Experience



All years of Experience



To filter salaries for Data Scientist, [Sign In](#) or [Register](#).

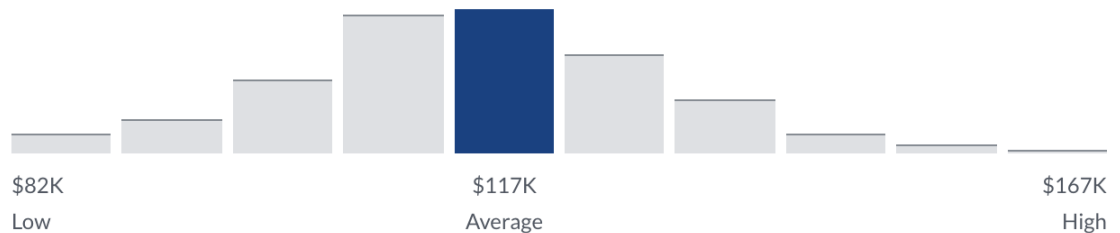


Very High Confidence

\$117,212 /yr

Average Base Pay

18,354 salaries



— 88,989

= 28,223!

Software systems for data analytics and ML over large and complex datasets are now critical for digital applications in many domains

The Age of “Big Data”/“Data Science”

The New York Times

SundayReview | NEWS ANALYSIS

The AgeForbes / Entrepreneurs

Forbes

By STEVE LOHR

MAR 25, 2015 @ 7:33 PM 4,407 VIEWS

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Drowning In Big Data - Finding Insight In A Digital

DATA

Data Scientist: The Sexiest Job of the 21st Century

Josh Steimle, CONT

by Thomas H. Davenport and D.J. Patil

FROM THE OCTOBER 2012 ISSUE

For roughly a decade, information about Big Data. The IDC industry will expect by 2018. What this

SUMMARY SAVE SHARE COMMENT 5 TEXT SIZE PRINT \$8.95 BUY COPIES

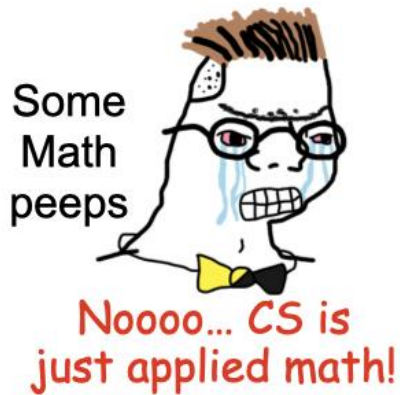


Harvard Business Review

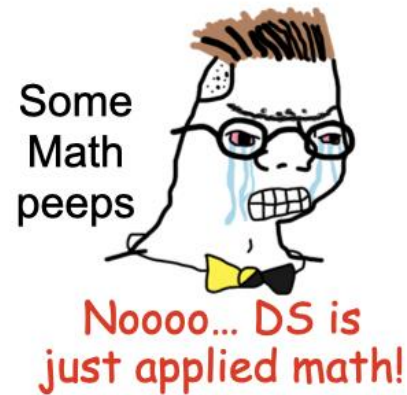
When Jonathan Goldman arrived for work in June 2006 at LinkedIn, the business networking site, the place still felt like a start-up. The company had just under 8 million accounts, and the number was growing quickly as existing members invited their friends and colleagues to join. But users weren't seeking out connections with the people who were already on the site at the rate executives had expected. Something was apparently missing in the social experience. As one LinkedIn manager put it, "It was like arriving at a conference reception and realizing you don't know anyone. So you just stand in the corner sipping your drink—

Meme from Previous DSC 102

THEN (1980s)



NOW (2020s)



15-213/15-513/14-513 Introduction to Computer Systems (ICS)

Fall 2023

- 15-213 Pittsburgh: Tue, Thu 12:30 PM–01:50 PM, GHC 4401, [Brian Railing](#) and [Phillip Gibbons](#)
- 14-513 Pittsburgh: Tue, Thu 12:30 PM–01:50 PM, CIC 1202, [David Varodayan](#)

12 units

The ICS course provides a programmer's view of how computer systems execute programs, store information, and communicate. It enables students to become more effective programmers, especially in dealing with issues of performance, portability and robustness. It also serves as a foundation for courses on compilers, networks, operating systems, and computer architecture, where a deeper understanding of systems-level issues is required. Topics covered include: machine-level code and its generation by optimizing compilers, performance evaluation and optimization, computer arithmetic, memory organization and management, networking technology and protocols, and supporting concurrent computation.

[Course Syllabus](#)

Prerequisites: 15-122

The ICS course provides a **programmer's view** of how computer systems execute programs, store information, and communicate. It enables students to become **more effective programmers**, especially in dealing with issues of performance, portability and robustness.

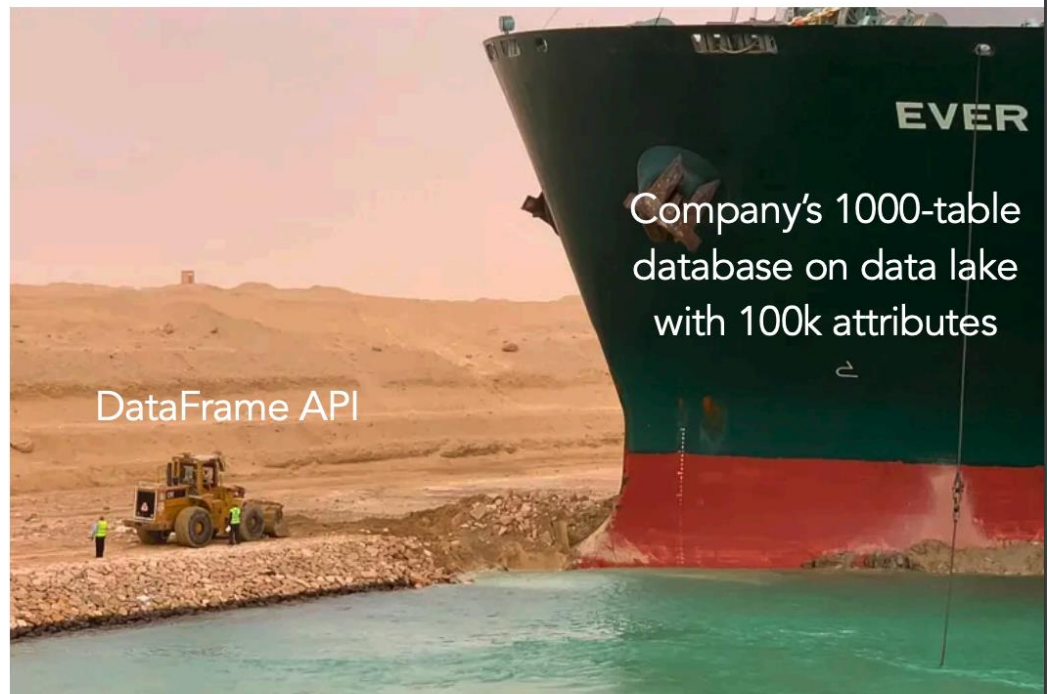
Vision

Data science professionals ought to be familiarized with data systems from a user's standpoint, as opposed to the conventional approach of a system implementer.

DSC 204a

DSC 204a Scalable Data Systems

- Haojian Jin



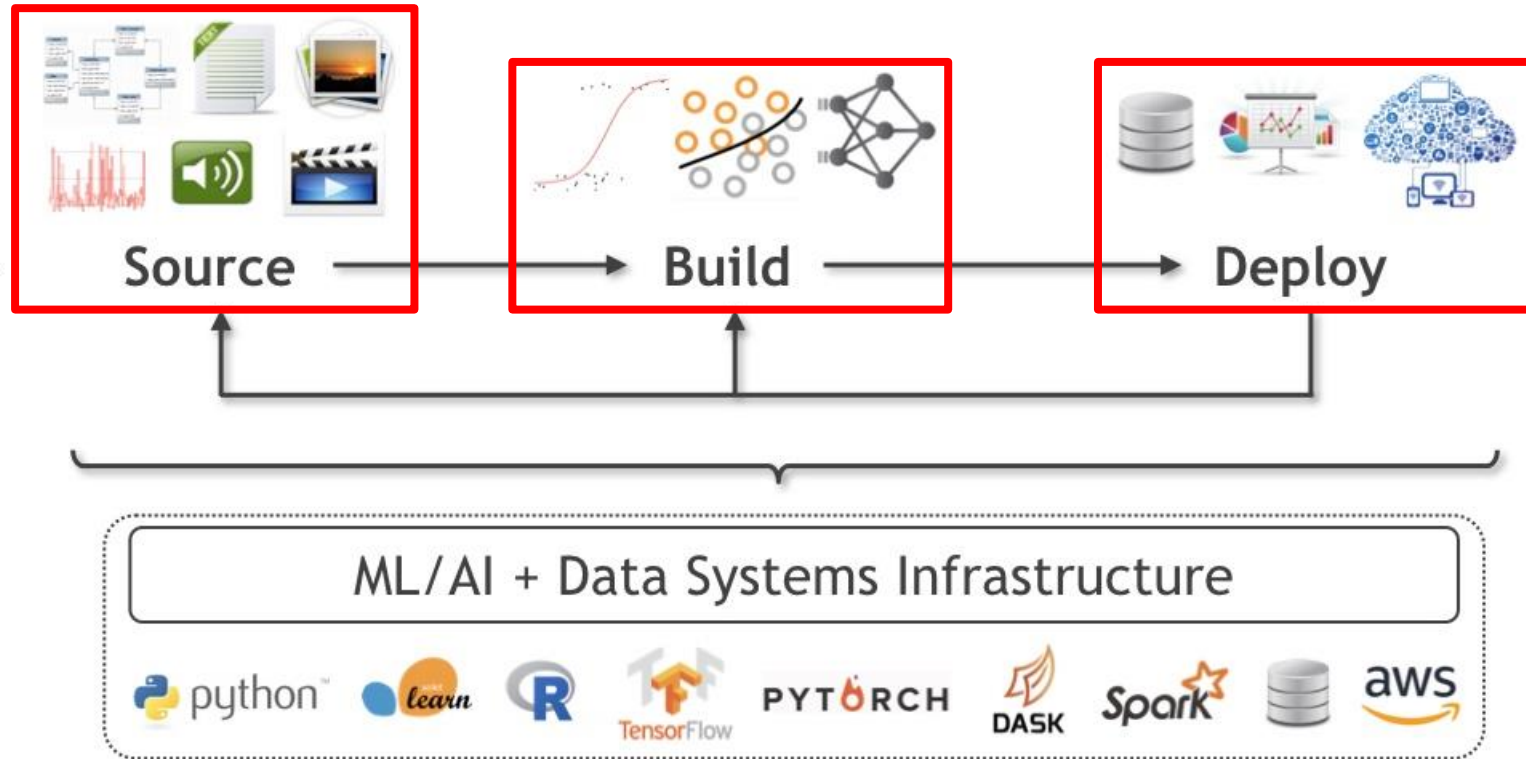
DSC 102 will get you thinking about the fundamentals of systems for scalable analytics

1. **“Systems”**: What resources does a computer have? How to store and efficiently compute over large data? What is cloud?
2. **“Scalability”**: How to scale and parallelize data-intensive computations?
3. **For “Analytics”**:
 1. **Source**: Data acquisition & preparation for ML
 2. **Build**: Model selection & deep learning systems
 3. **Deploying** ML models
4. Hands-on experience with scalable analytics tools

The Lifecycle of ML-based Analytics



Data Scientist/
ML Engineer



ML Systems

Q: What is a Machine Learning (ML) System?

- ❖ A data processing system (aka *data system*) for mathematically advanced data analysis operations (inferential or predictive):
 - ❖ Statistical analysis; ML, deep learning (DL); data mining (domain-specific applied ML + feature eng.)
 - ❖ *High-level APIs* to express ML computations over (large) datasets
 - ❖ *Execution engine* to run ML computations efficiently

Categorizing ML Systems

❖ Orthogonal Dimensions of Categorization:

1. **Scalability:** In-memory libraries v. Scalable ML system (works on larger-than-memory datasets)
2. **Target Workloads:** General ML library v. Decision tree-oriented v. Deep learning, etc.
3. **Implementation Reuse:** Layered on top of scalable data system v. Custom from-scratch framework

Major Existing ML Systems

General ML libraries:

In-memory:



Disk-based files:



Layered on RDBMS/Spark:



Cloud-native:



“AutoML” platforms:



Decision tree-oriented:



Microsoft
LightGBM

Deep learning-oriented:



PYTORCH

Data Systems Concerns in ML

Key concerns in ML:

Q: How do “ML Systems” relate to ML?

Runtime efficiency (sometimes)

Additional key *practical* concerns in ML Systems:

ML Systems : ML :: Computer Systems : TCS

Usability

Manageability

Developability

*Long-standing
concerns in the
DB systems
world!*

Q. Q. What do you think is the most effective way to deal with the high level of corruption in Russia?

Conceptual System Stack Analogy

Relational DB Systems

ML Systems

Theory

First-Order Logic
Complexity Theory

Learning Theory
Optimization Theory

Program Formalism

Relational Algebra

Tensor Algebra
Gradient Descent

Program Specification

SQL

TensorFlow?
Scikit-learn?

Program Modification

Query Optimization

???

Execution Primitives

Parallel Relational
Operator Dataflows

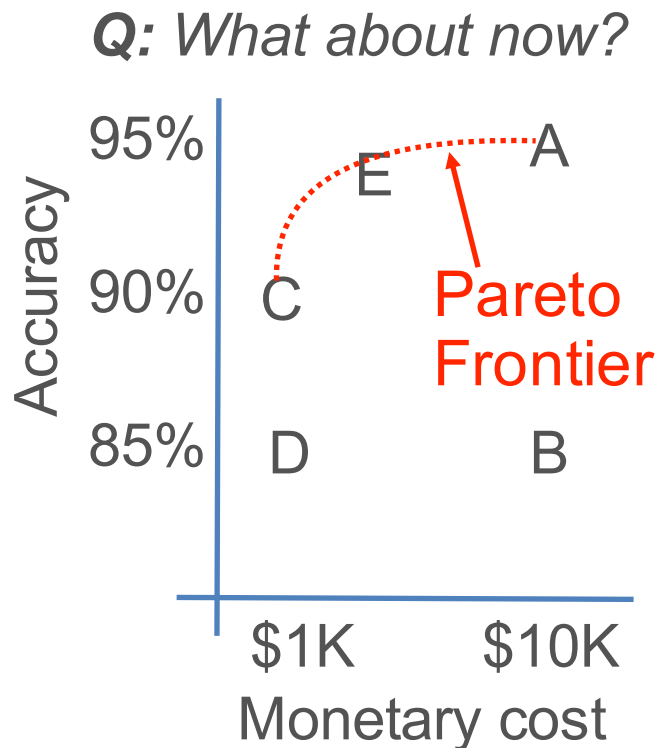
Depends on ML Algorithm

Hardware

CPU, GPU, FPGA, NVM, RDMA, etc.

Real-World ML: Pareto Surfaces

Q: Suppose you are given ad click-through prediction models A, B, C, and D with accuracies of 95%, 85%, 90%, and 85%, respectively. Which one will you pick?



- ❖ Real-world ML users must grapple with multi-dimensional *Pareto surfaces*: accuracy, monetary cost, training time, scalability, inference latency, tool availability, interpretability, fairness, etc.
- ❖ *Multi-objective optimization* criteria set by application needs / business policies.

Learning Outcomes of this course

- ❖ **Explain** the basic principles of the memory hierarchy, parallelism paradigms, scalable data systems, and cloud computing.
- ❖ **Identify** the abstract data access patterns of, and opportunities for parallelism and efficiency gains in, data processing and ML algorithms at scale.
- ❖ **Outline** how to use cluster and cloud services, dataflow (“Big Data”) programming with MapReduce and Spark, and ML tools at scale.
- ❖ **Apply** the above programming skills to create end-to-end pipelines for data preparation, feature engineering, and model selection on large-scale datasets.
- ❖ **Reason** critically about practical tradeoffs between accuracy, runtimes, scalability, usability, and total cost.

What this course is NOT about

- ❖ NOT a course on databases, relational model, or SQL
 - ❖ Take DSC 100 instead (pre-requisite)
- ❖ NOT a course on internal details of RDBMSs
 - ❖ Take CSE 132C instead
- ❖ NOT a training module for how to use Spark
- ❖ NOT a course on ML or data mining *algorithmics*; instead, we focus on ML *systems*

Now for the course logistics ...

Prerequisites

- ❖ **DSC 100** (or equivalent) is necessary
- ❖ Transitively **DSC 80**; a mainstream ML algorithmics course is necessary
- ❖ Proficiency in Python programming
- ❖ For all other cases, email me with proper justification; a waiver can be considered

Grading Scheme

Hybrid of relative and absolute; grade is better of the two

Grade	Relative Bin (Use strictest)	Absolute Cutoff ($\geq$)
A+	Highest 5%	95
A	Next 10% (5-15)	90
A-	Next 15% (15-30)	85
B+	Next 15% (30-45)	80
B	Next 15% (45-60)	75
B-	Next 15% (60-75)	70
C+	Next 5% (75-80)	65
C	Next 5% (80-85)	60
C-	Next 5% (85-90)	55
D	Next 5% (90-95)	50

Example: Score 82 but 33%ile; Rel.: B-; Abs.: B+; so, B+

General Dos and Do NOTs

Do:

- ❖ Follow all announcements on Piazza
- ❖ Try to join the lectures/discussions live
- ❖ Raise your hand before speaking
- ❖ View/review podcast videos asynchronously by yourself
- ❖ To contact me/TAs, use private Slack; if you really need to email, use “DSC 102:” as subject prefix

Do NOT:

- ❖ Harass, intimidate, or intentionally talk over others
- ❖ Violate academic integrity on the PAs, exams, or other components; I am *very strict* on this matter!

Reasonable person.

- (1) Everyone will be reasonable.
- (2) Everyone expects everyone else to be reasonable.
- (3) No one is special.
- (4) Do not be offended if someone suggests you are not being reasonable.